Project Design Phase
Proposed Solution Template

Date	18 February 2026
Team ID	LTVIP2026TMIDS65644
Project Name	Rising Waters: A Machine Learning Approach To Flood Prediction
Maximum Marks	2 Marks

Proposed Solution – Rising Waters: A Machine Learning Approach To Flood Prediction

The Rising Waters project proposes a data-driven solution to predict flood risk levels using machine learning techniques. By leveraging historical weather data—such as rainfall intensity, temperature, humidity, and river water levels—the system trains a predictive model capable of classifying flood risk into categories like Low, Medium, or High. Logistic Regression or other suitable algorithms are applied to identify patterns and correlations between climatic factors and past flood events. The model is integrated into a user-friendly web interface built with Flask, allowing users to input current weather parameters and instantly receive a flood risk assessment. This approach enables early warnings, supports disaster preparedness, and aids authorities in taking timely preventive measures, ultimately reducing potential damage and safeguarding communities.